Section 4.2- Virtual Cell

Building World Model and Simulating Data for Molecular Medicine

2026 Spring

[LLM Agents Foundation & Applications]

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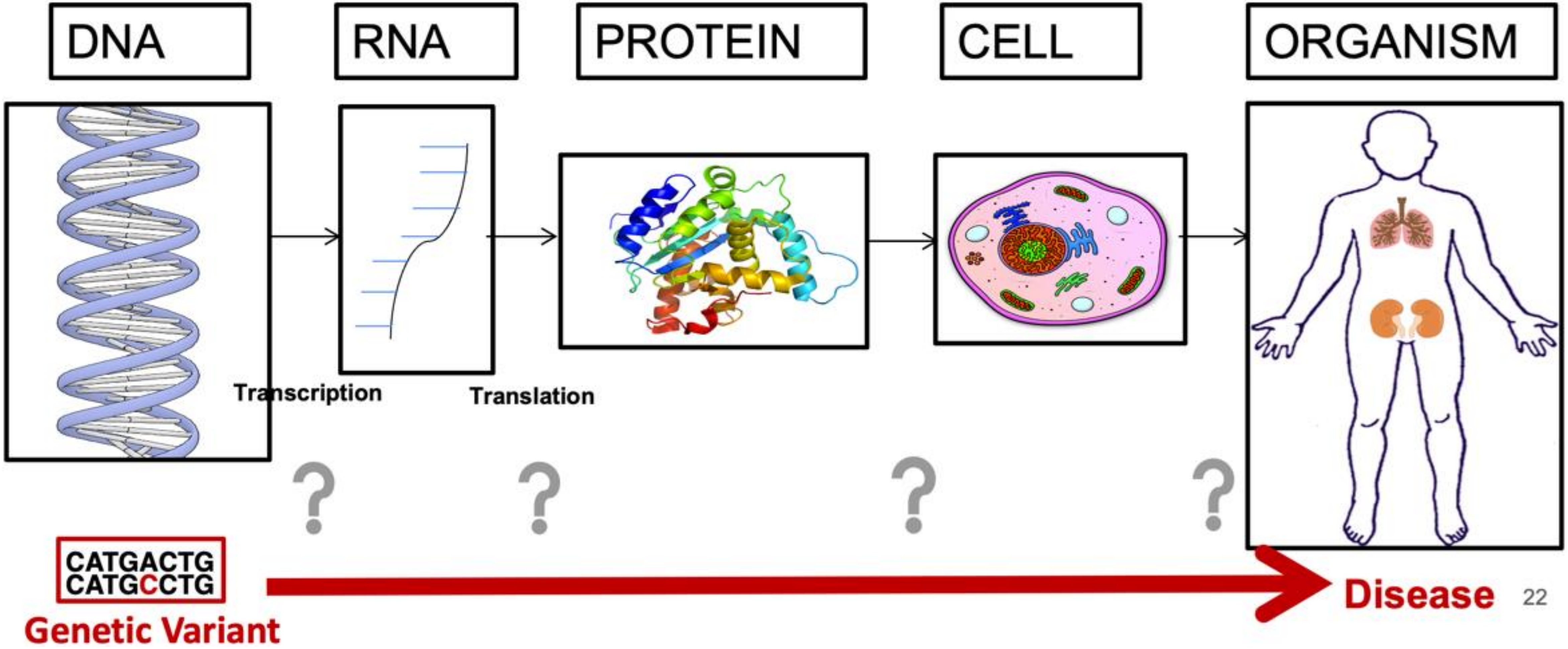
20260210

Biology in a Slide:



For Predictive Biology

THE GOAL

To build 'Virtual Cells'—AI models capable of predicting cellular responses to perturbations indistinguishably from

THE BOTTLENECK

Progress is stalled by a lack of standardized evaluations and high-quality, reproducible ground-truth data.

THE INITIATIVE

The Virtual Cell Challenge (VCC)—an annual, open-source competition launching in 2025 to evaluate model generalization.

THE ASSET

Powered by a purpose-built, high-depth H1 hESC dataset generated specifically for this challenge by the Arc Institute.

How to Build the AI Virtual Cell with Artificial Intelligence

How to Build the Virtual Cell with Artificial Intelligence: Priorities and Opportunities

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Why Model the Cell?

- The cell = fundamental unit of life
- Disease emerges from cellular dysfunction
- Current models:
  - Rule-based
  - Mechanistic but narrow
  - Hard to scale across modalities

Problem: Biology is multiscale, nonlinear, massively interactive

Limitations of Traditional Whole-Cell Models

- Differential equations
- Stochastic simulations
- Agent-based models

Work well for:

- Specific pathways
- Bacterial organisms

Struggle with:

- Multiscale integration
- Human cellular complexity
- Cross-modality reasoning

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Omics Revolution

- Single-cell RNA-seq
- Spatial transcriptomics
- Imaging
- Multi-modal atlases

AI Revolution

- Foundation models
- Transformers
- Diffusion models
- Generative modeling

Vision: The AI Virtual Cell (AIVC)

A learned, multi-scale, multi-modal neural simulator of cells and tissues

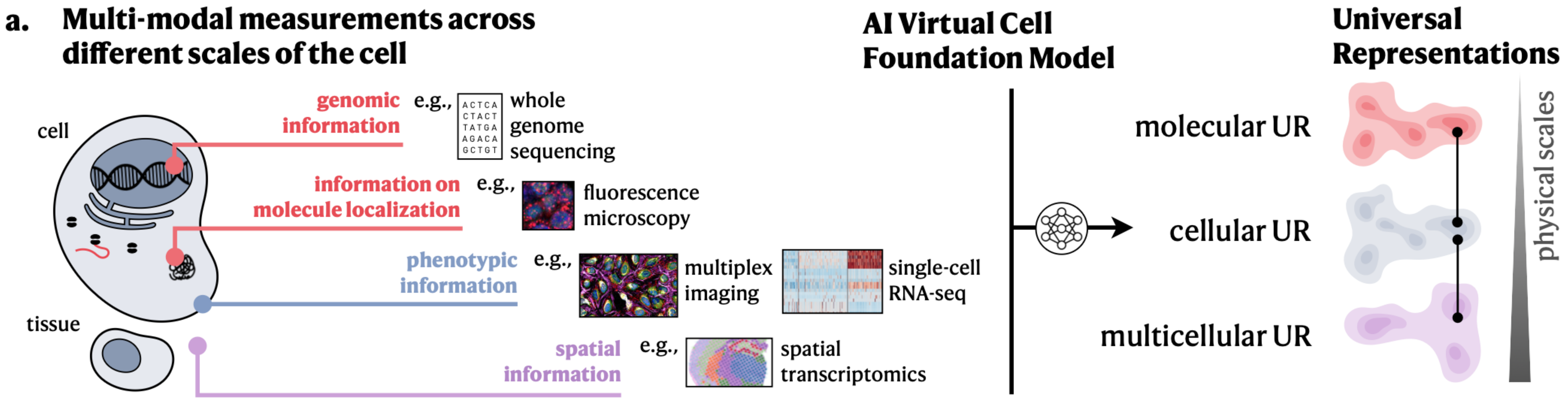
Core capabilities:

1. Universal biological representations
2. Predict behavior and dynamics
3. Enable in silico experimentation

Universal Representations (URs)

Must generalize to unseen states

- Embedding of biological state
- Integrates:
  - Species
  - Modalities
  - Scales
  - Conditions



Three Physical Scales

1. Molecular UR
2. Cellular UR
3. Multicellular UR

Each scale builds on the previous

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Molecular Scale

Data:

- DNA
- RNA
- Proteins
- Metabolites

Modeling:

- Sequence transformers
- Language modeling
- Atomic-level models

Goal: Learn molecular grammar

Three Physical Scales

1. Molecular UR
2. Cellular UR
3. Multicellular UR

Each scale builds on the previous

Cellular Scale

Data:

- scRNA-seq
- scATAC-seq
- Proteomics
- Imaging

Modeling:

- Multi-modal transformers
- CNNs / Vision Transformers
- Cross-modal integration

Output: Unified cell-state embedding

Three Physical Scales

1. Molecular UR
2. Cellular UR
3. Multicellular UR

Each scale builds on the previous

Multicellular Scale

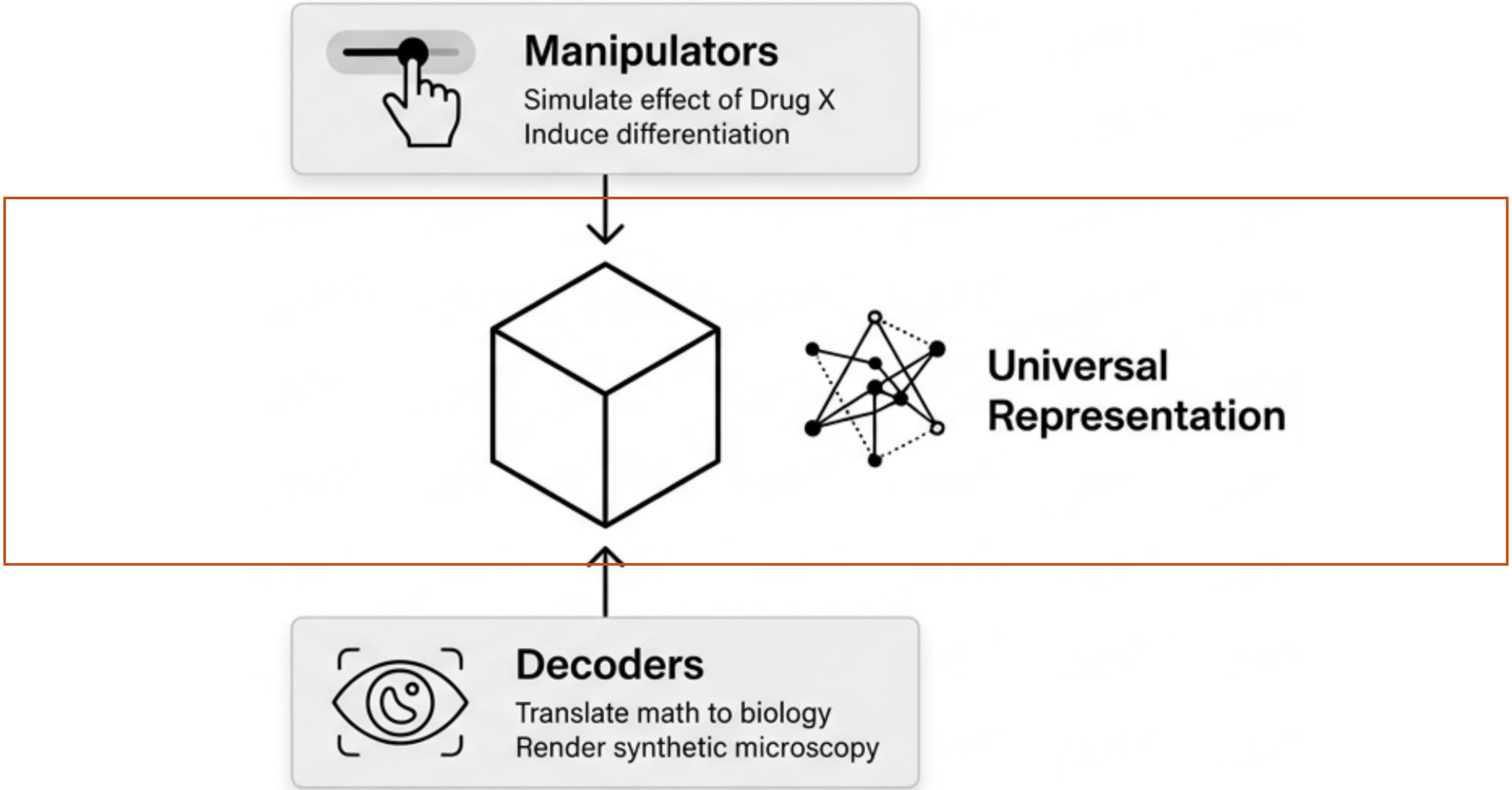
Data:

- Spatial transcriptomics
- Tissue imaging
- Spatial proteomics

Modeling:

- Graph neural networks
- Equivariant networks
- Spatial transformers

Goal: Model cell-cell interactions and niches


Manipulators
Simulate effect of Drug X
Induce differentiation
Universal
Representation
Decoders
Translate math to biology
Render synthetic microscopy

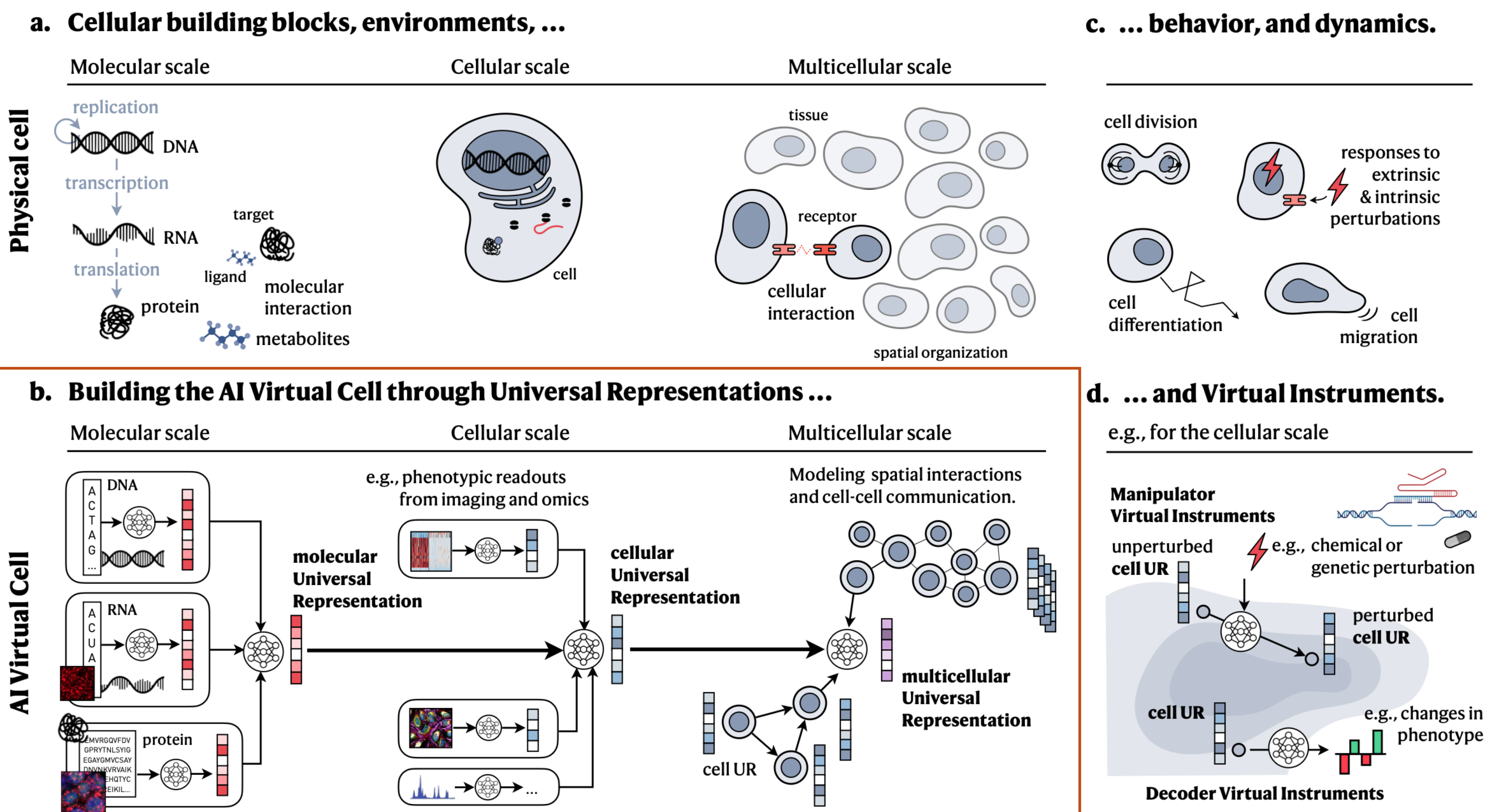


Figure 2: Overview of the AI Virtual Cell. a. Similar to biological cells, **b.** the AI Virtual Cell models cell biology across different physical scales, including molecular, cellular, and multicellular. Along the physical dimension, the first scale models the state and interactions of individual molecules, such as those of the central dogma, as well as additional molecules like metabolites. Molecules

Data Requirements

Must capture:

- Multi-scale biology
- Multi-modal signals
- Perturbations
- Temporal dynamics
- Human diversity

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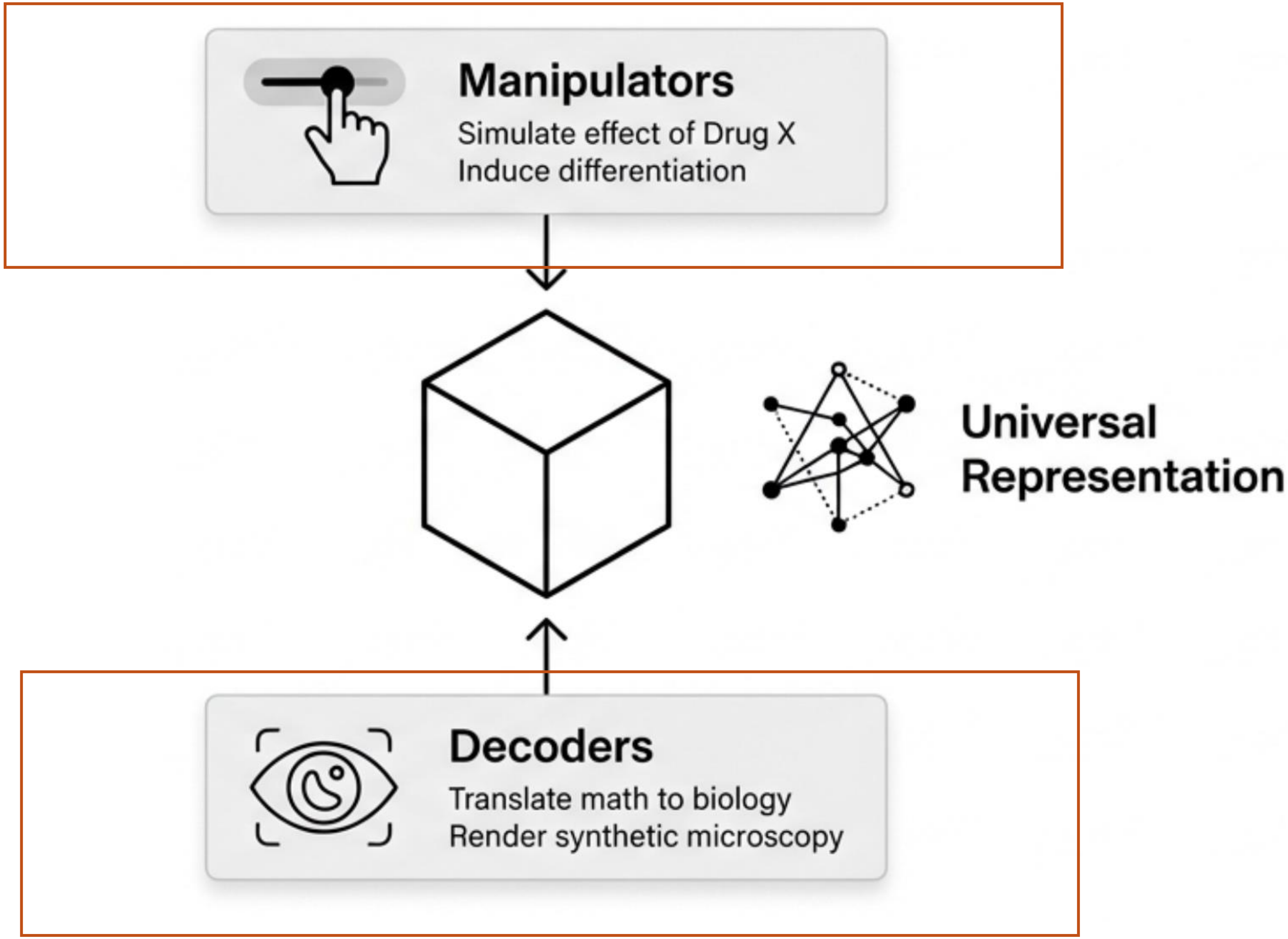
Data Challenges

- Species bias
- Population bias
- Technical noise vs biological variation
- Combinatorial explosion
- Unknown scaling laws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Virtual instruments are neural networks that manipulate or decode these representations.

Virtual Instruments (VIs)

Decoder VIs

- UR → phenotype / image / label

Manipulator VIs

- UR → modified UR
- Simulate perturbations

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Modeling Dynamics with Manipulators

Predict:

- Differentiation
- Migration
- Disease progression
- Perturbation response

Methods:

- Diffusion models
- Autoregressive transformers
- Flow matching

AI Architectures: Transformers

Used for:

- DNA
- RNA
- Protein sequences
- scRNA-seq

Strength:

- Self-attention models gene interactions

CNNs & Vision Transformers

Used for:

- Microscopy
- Histology
- Spatial phenotyping

Strength:

- Spatial pattern recognition

Diffusion Models

Used for:

- Dynamic modeling
- Continuous trajectories
- Perturbation prediction

Strength:

- High-dimensional generative modeling

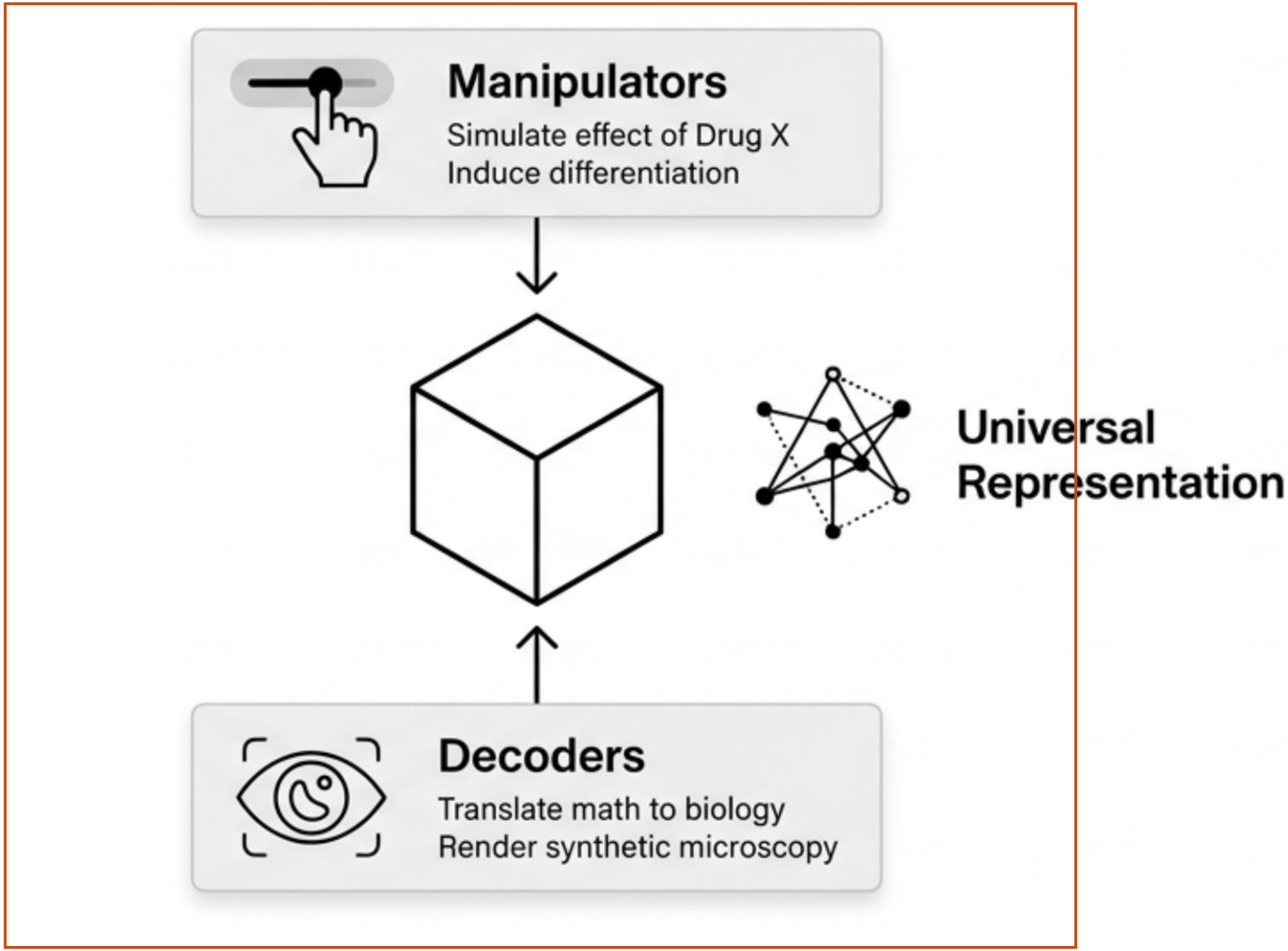
Graph Neural Networks

Used for:

- Tissue organization
- Protein structures
- Cell-cell communication

Strength:

- Structured relational modeling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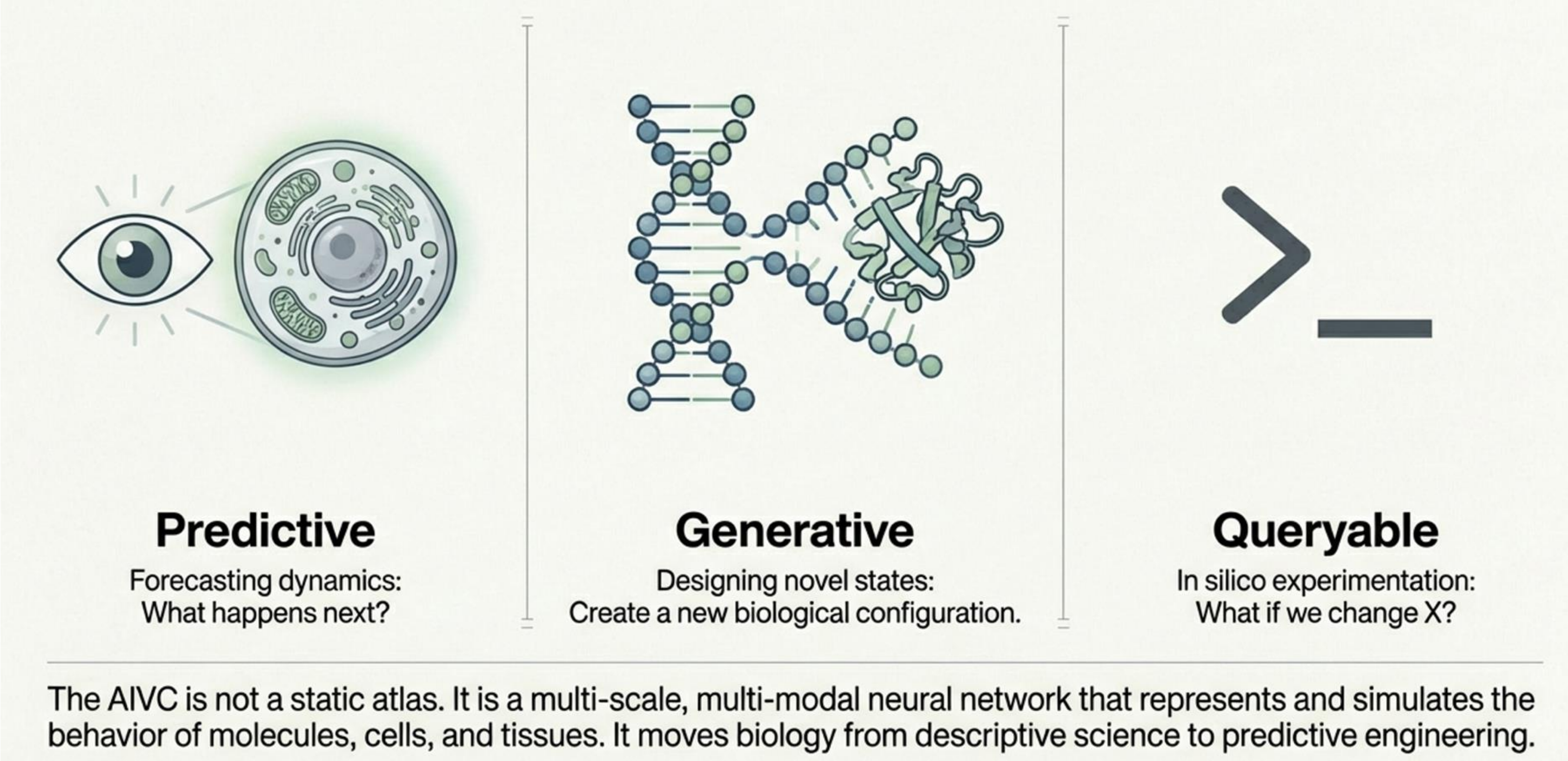
Outlook



AI Virtual Cells could:

- Redefine biomedical research
- Enable programmable biology
- Transform drug discovery
- Create digital biological twins

A paradigm shift from modeling fragments → modeling life systems

Outlook

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- Redefine biomedical research
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A paradigm shift from modeling fragments → modeling life systems

In Silico Experimentation

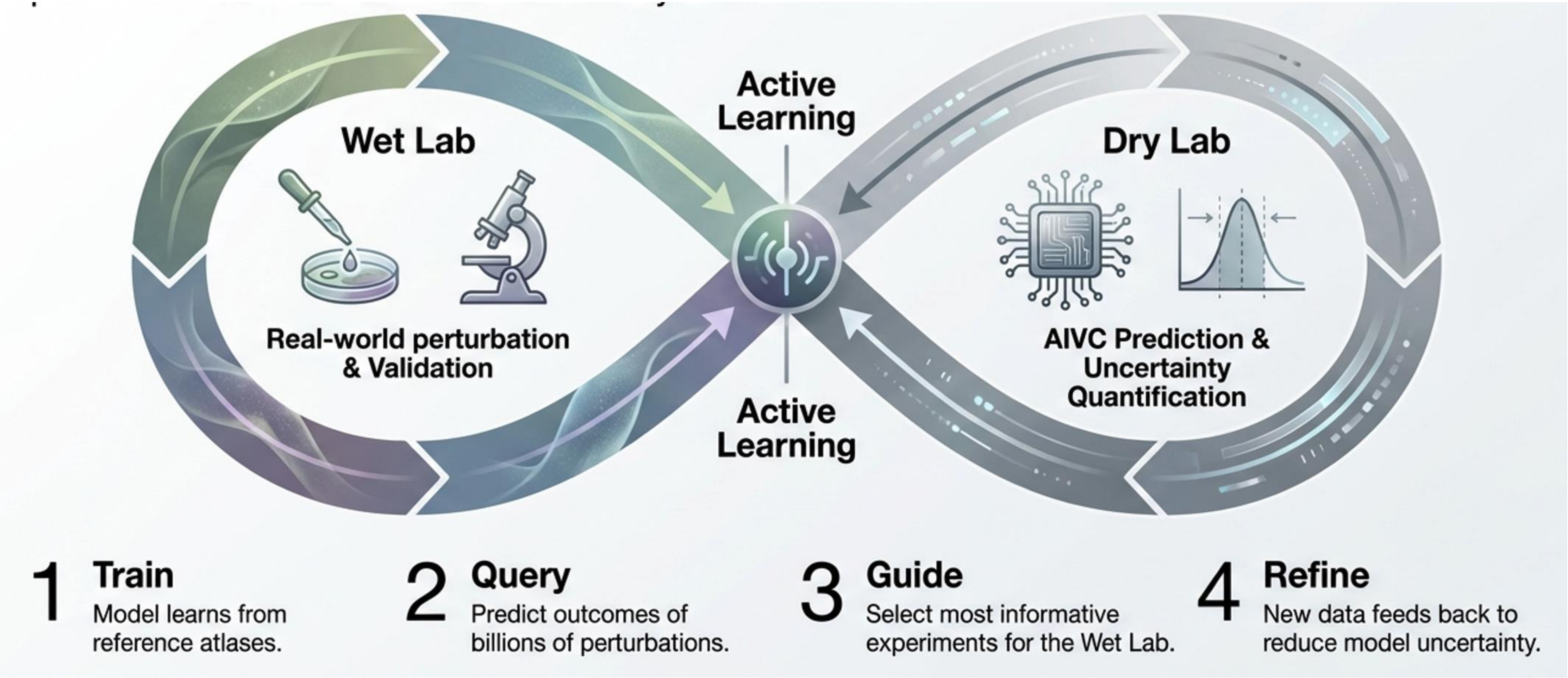
AIVC can:

- Screen combinatorial perturbations
- Simulate expensive assays
- Explore impossible experiments
- Guide real experiments

Creates a **lab-in-the-loop system**

Enable Lab in the loop

Lab-in-the-loop refers to an **active and iterative process** that creates a dynamic interaction between physical laboratory experimentation and AI-driven computational modeling. In this framework, the AI Virtual Cell (AIVC) acts as a "virtual laboratory" that facilitates a seamless interface with real-world experiments



Evaluating an AIVC

Metrics:

- Generalization to unseen contexts
- Cross-modal reconstruction
- OOD robustness
- Hypothesis generation ability

Beyond Accuracy: Biological Insight

AIVC should:

- Propose testable hypotheses
- Suggest causal mechanisms
- Link molecular changes → phenotype
- Reduce hypothesis search space

Application 1: Phenotypic Drug Discovery

- Simulate disease context
- Virtual phenotypic screens
- Prioritize interventions
- Accelerate cell-based therapies

Application 2: Spatial Oncology

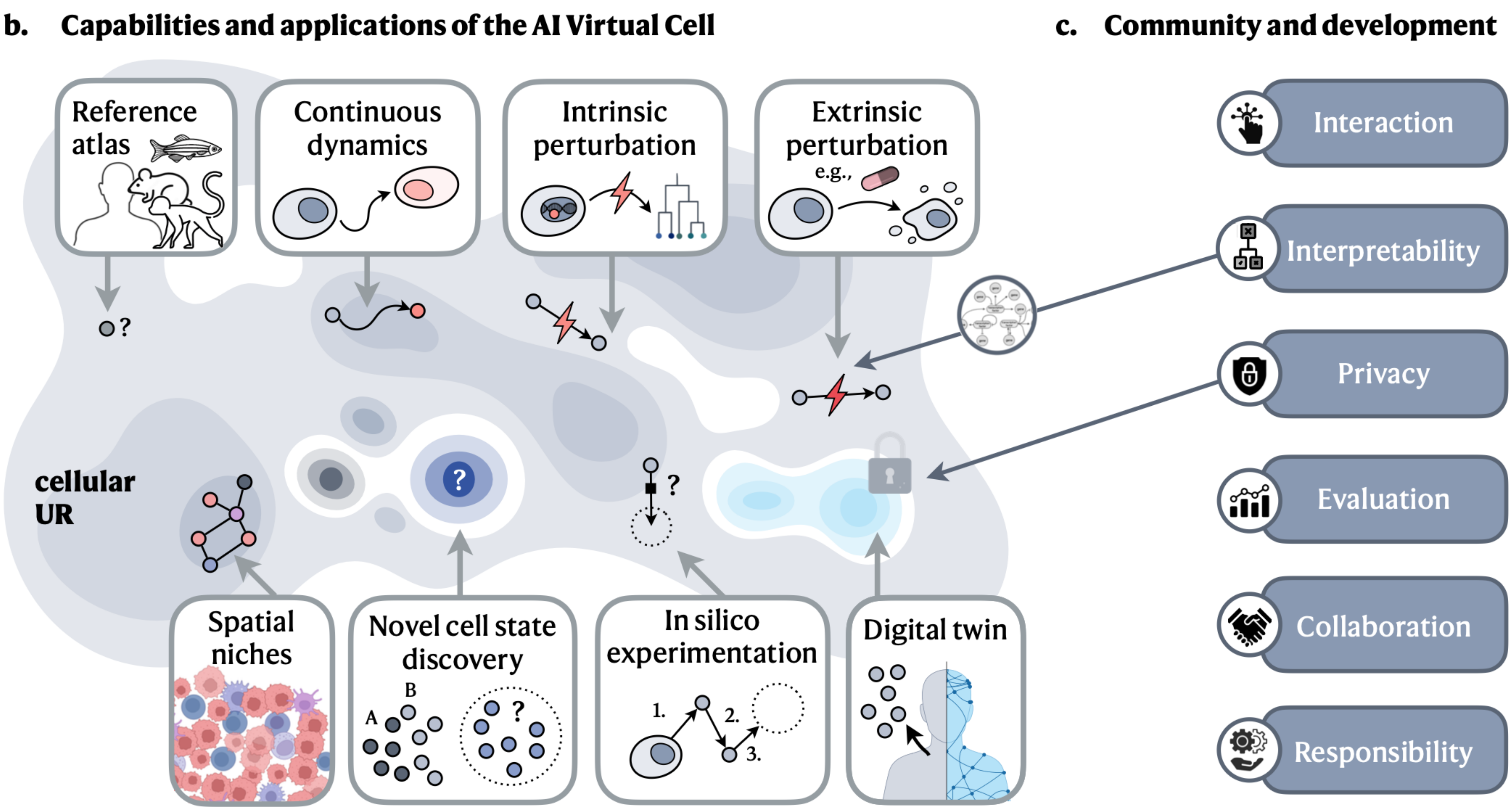
- Model tumor microenvironment
- Identify pan-cancer niches
- Discover resistance mechanisms
- Personalize treatment strategies

Application 3: Digital Twin Patients

- Patient-specific UR
- Integrate genomics + imaging + clinical data
- Predict disease progression
- Guide personalized therapy

Application 4: Hypothesis-Generating Engine

- Shift from validation → generation
- Active learning loop
- Toward self-driving biology labs



re 1: **Capabilities of the AI Virtual Cell. a.** The AI Virtual Cell provides a Universal Representation of a cell state that can be

Why Now?

- Massive cell atlases
- Biological foundation models
- Generative AI advances
- Cross-sector collaboration

Grand Challenges

1. Define core capabilities
2. Ensure self-consistency across scales
3. Balance interpretability vs utility
4. Prioritize data collection
5. Ensure ethical and equitable use
6. Build open collaborative infrastructure

Open questions:

NEXT Paper

- What is the minimal viable AIVC?
- How do we benchmark across scales?
- What inductive biases are essential?
- Should AIVCs be centralized or federated?

Virtual Cell Challenge

Toward a Turing Test for the Virtual Cell

Roohani, Hua, Tung et al. · Arc Institute · Cell, June 2025

What Is a "Virtual Cell"?

> A **virtual cell** is a computational model that learns the relationship between cell state and function, and predicts the consequences of perturbations—such as gene knockdowns or drug applications—across cell types and contexts.

Why does this matter?

- Understanding how cells respond to perturbations is central to basic biology and drug discovery
- Experimental testing of all perturbations × all cell types is **impossibly expensive**
- A model that can generalize across contexts could revolutionize biology and medicine

Why Do We Need a Benchmark for Virtual Cells?

Why Now? Conditions Have Changed

- Massively improved single-cell measurement technologies (scRNA-seq, CRISPRi screens)
- Richer perturbational datasets — genome-wide Perturb-seq at scale
- Advances in machine learning and foundation models (scGPT, GEARS)

Inspiration: CASP for Protein Folding

- CASP (1994–) created a recurring blind competition for protein structure prediction
- It catalyzed the field and ultimately enabled AlphaFold2 — which solved protein folding
- ImageNet catalyzed computer vision in the same way
- **The Virtual Cell Challenge aims to do the same for predictive cell biology**

But no standardized benchmarks exist to evaluate whether models capture generalizable biological structure vs. dataset-specific patterns.

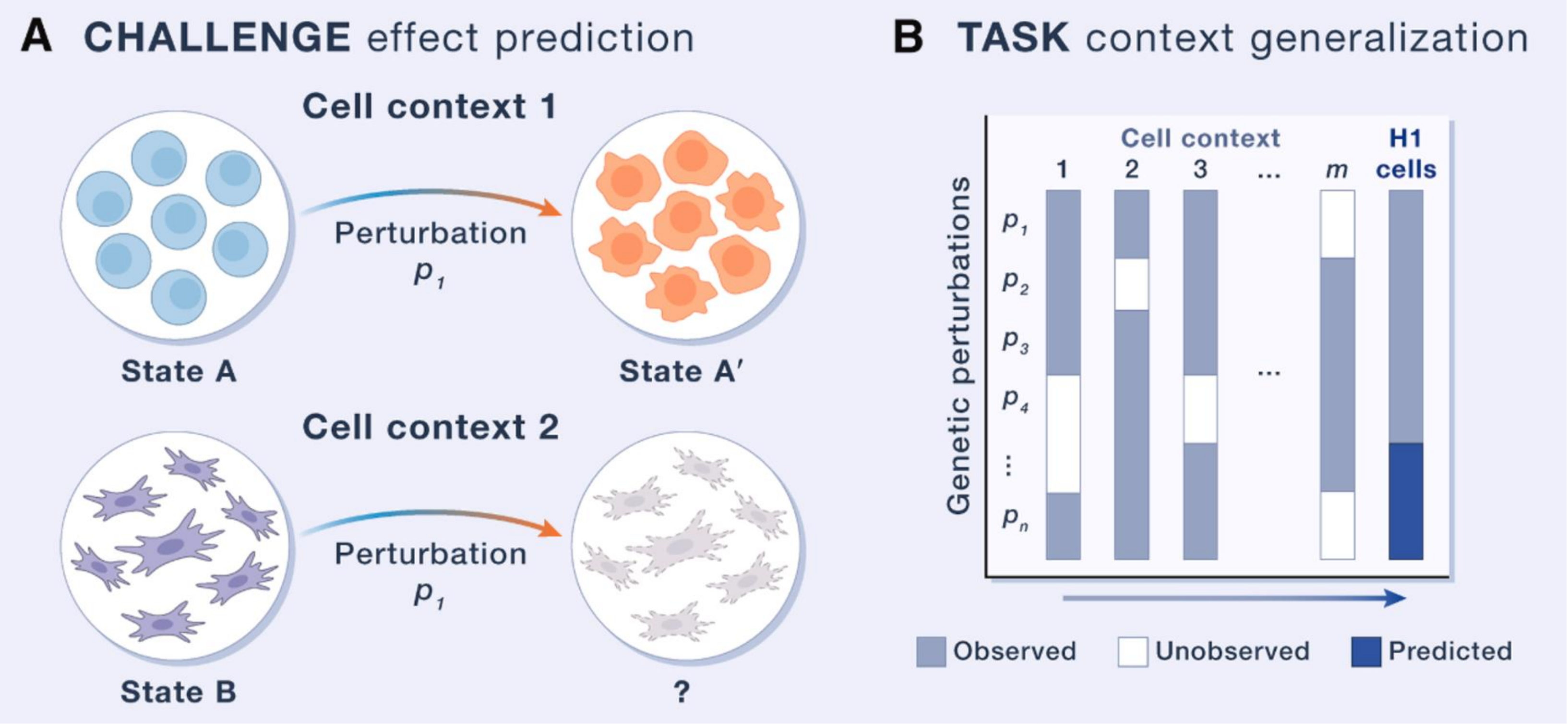
Past Competitions and Remaining Gaps

Competition	What It Tested	Gap
Broad Cancer Immunotherapy Challenge	Broad phenotypic shifts in T cells	No gene expression resolution
NeurIPS-Kaggle 2023	Gene expression response to **small molecules**	Limited to chemicals; no genetic perturbations
Virtual Cell Challenge (2025)	Gene expression response to **genetic perturbations**	← *This work*

Why genetic perturbations?

- Precisely defined targets → ideal for probing causal gene-function relationships
- Transcriptional changes can be more subtle → harder to predict → better test

Task: Context Generalization



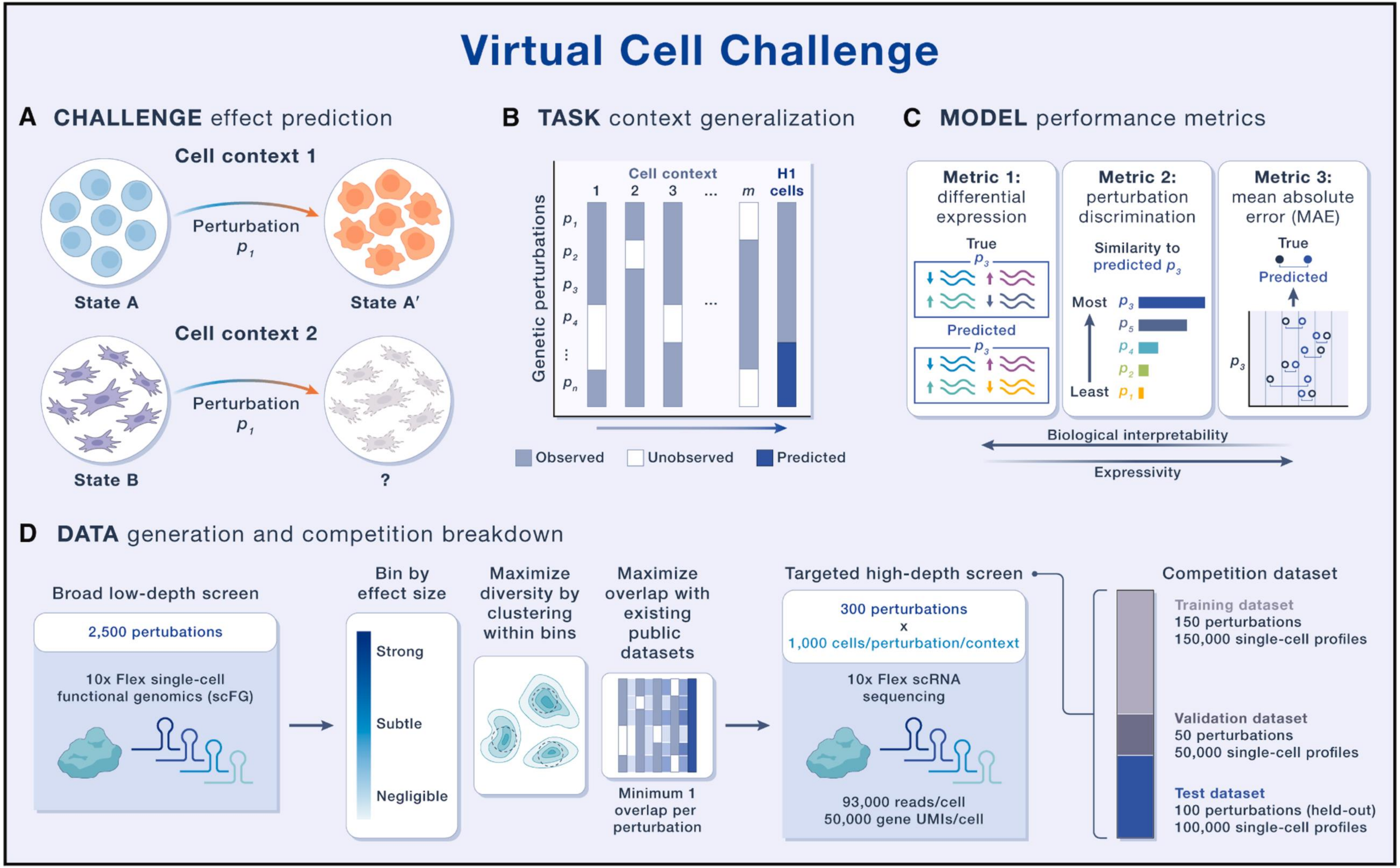
The key challenge: generalization across cell types

Task formulation:

- Participants have seen perturbation effects in **multiple other cell contexts**
- A **subset** of perturbation effects in H1 hESCs is provided (few-shot adaptation)
- Goal: predict the effects of **held-out perturbations** in H1 hESCs

Why few-shot, not zero-shot?

> Most published datasets span only a handful of cell lines — true zero-shot generalization to new cell states is likely premature at this stage.



Virtual Cell Challenge
A CHALLENGE effect prediction
Cell context 1
Perturbation p_1
State A
State A′
Cell context 2
Perturbation p_1
State B
?
B TASK context generalization
Cell context
1 2 3 ... m
H1 cells
Genetic perturbations
p_1 p_2 p_3 p_4 ⋮ p_n
Observed
Unobserved
Predicted
C MODEL performance metrics
Metric 1: differential expression
True p_3
Predicted p_3
Metric 2: perturbation discrimination
Similarity to predicted p_3
Most
Least
p_3 p_5 p_4 p_2 p_1
Metric 3: mean absolute error (MAE)
True
Predicted
p_3
Biological interpretability
Expressivity
D DATA generation and competition breakdown
Broad low-depth screen
2,500 pertubations
10x Flex single-cell functional genomics (scFG)
Bin by effect size
Strong
Subtle
Negligible
Maximize diversity by clustering within bins
Maximize overlap with existing public datasets
Minimum 1 overlap per perturbation
Targeted high-depth screen
300 perturbations x 1,000 cells/perturbation/context
10x Flex scRNA sequencing
93,000 reads/cell
50,000 gene UMIs/cell
Competition dataset
Training dataset
150 perturbations
150,000 single-cell profiles
Validation dataset
50 perturbations
50,000 single-cell profiles
Test dataset
100 perturbations (held-out)
100,000 single-cell profiles

Task Design: core components & Three Metrics

Task: Predict Gene Expression Response to CRISPRi Perturbations in H1 hESCs

Core components of the challenge:

- Annual, open competition hosted by Arc Institute
- Purpose-built high-quality benchmark dataset (H1 hESC line)
- Public leaderboard for real-time progress tracking
- Three complementary evaluation metrics
- Training data from the Arc Virtual Cell Atlas + public datasets

Website: virtualcellchallenge.org

The Benchmark in Context

What good performance would demonstrate:

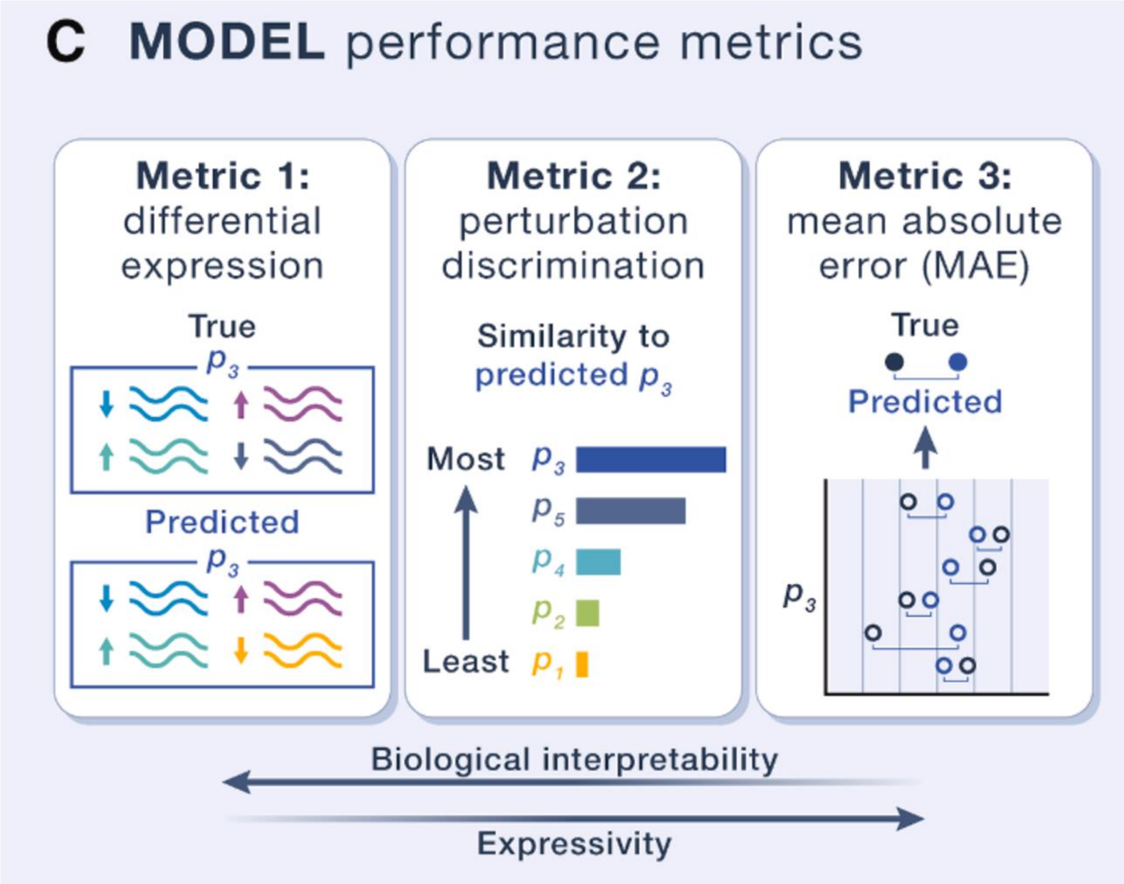
- A model can generalize gene expression predictions to a new, unseen cell type (H1 hESC)
- Using only a **few-shot subset** of H1 hESC perturbations for adaptation
- Achieving balanced scores on all three complementary metrics
- Reproducing biologically meaningful differential expression signatures

Key Quality Metrics Achieved

Parameter	Value
Total cells profiled	~300,000
Perturbations profiled	300 genes (CRISPRi)
Median cells / perturbation	~1,000 cells
Mean reads per cell	~93,000
Median gene UMIs per cell	~50,000

The Three Evaluation Metrics

Combined score weighted across all three metrics. Minimum thresholds enforced on all metrics — prevents gaming by models that excel at one dimension at the expense of others.



Metric 1
Differential Expression Score
How accurately does the model predict DE genes?

Metric 2
Perturbation Discrimination Score
Can the model distinguish between perturbations?

Metric 3
Mean Absolute Error (MAE)
Global accuracy across all genes

Virtual Cell Challenge

Benchmark Dataset: High-Quality CRISPRi Screen

Dataset Generation Pipeline

- Start: Broad low-depth screen of 2,500 CRISPRi perturbations in H1 hESCs
- Select 300 perturbations covering broad range of effect sizes (strong → subtle → negligible) and diversity of phenotypic responses (clustering-based selection)
- High-depth targeted sequencing using 10x Genomics Flex chemistry (fixed-cell, reduced batch effects)

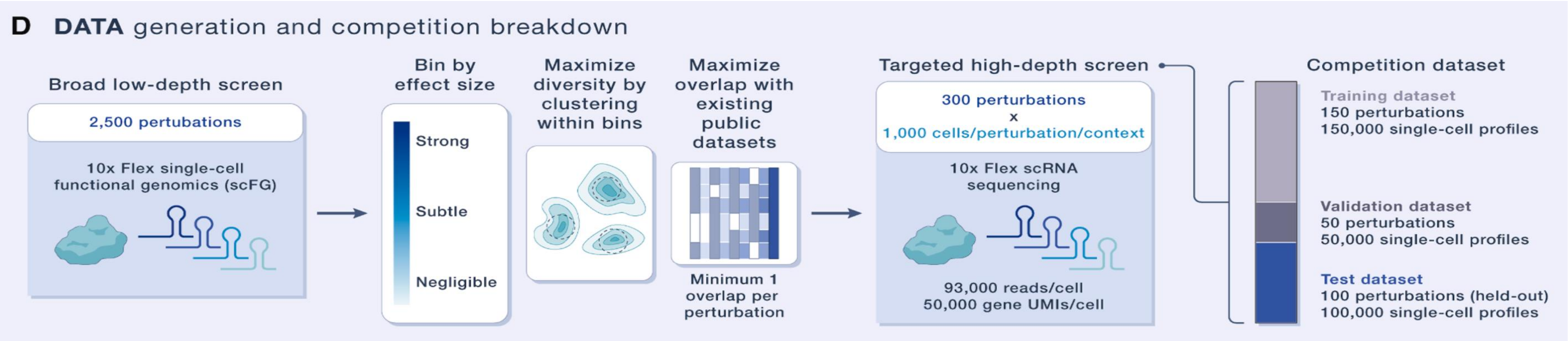
Competition Dataset Split

Training set

150 perturbations · ~150,000 cells · Released at launch

Validation set

50 perturbations · ~50,000 cells · Live leaderboard



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Additional Training Resources: Arc Virtual Cell Atlas — scBaseCount + Tahoe-100M (>100M cells, largest public perturbational scRNA-seq dataset). Combined: >350 million cells and counting.

Competition Dataset Split

Training set

150 perturbations · ~150,000 cells · Released at launch

Validation set

50 perturbations · ~50,000 cells · Live leaderboard

Test set

100 perturbations · ~100,000 cells · Released 1 week before deadline · Final rankings

Goals of the Virtual Cell Challenge

The challenge is designed to achieve three outcomes:

1. Surface best practices and data considerations

- Training set quality, sequencing depth, platform choice
- Guidelines for reproducible and principled data generation

2. Identify methodological bottlenecks

- Clarify which model and data choices drive generalization
- Converge on reproducible standards for perturbation modeling

3. Highlight the importance of data quality

- Cell coverage, sequencing depth, and technology choice
- By underscoring these issues, drive higher-quality community data generation

Future Directions

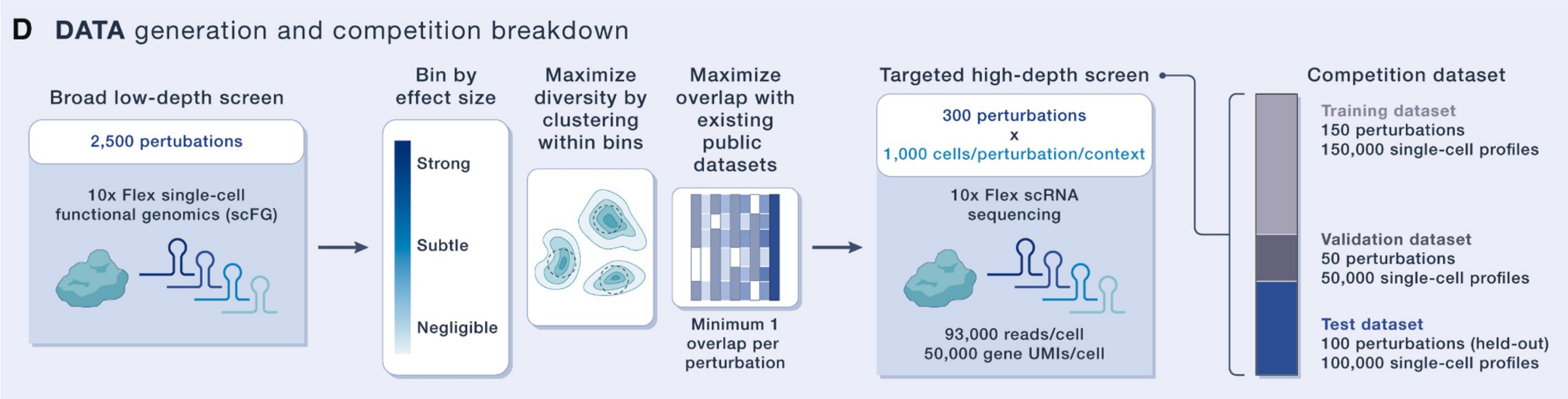
Expanding the challenge over time:

Near-term expansions:

- Combinatorial perturbations (gene × gene, gene × drug)
- Cross-cell-type generalization at scale
- Multi-modal predictions (proteomics, epigenomics)

Longer-term vision:

- Temporal and spatial dimensions
- Multicellular systems
- **Inverse design:** identify perturbations that achieve a desired cellular effect (therapeutic relevance)



The Broader Vision: A Turing Test for the Virtual Cell

> "Virtual cells are poised to become foundational tools for biology, and to ensure that they reach their potential, we need clear and rigorous evaluations."

Analogies:

- **CASP** → catalyzed protein structure prediction → AlphaFold
- **ImageNet** → catalyzed computer vision
- **Virtual Cell Challenge** → could catalyze predictive cell biology

The ultimate goal:

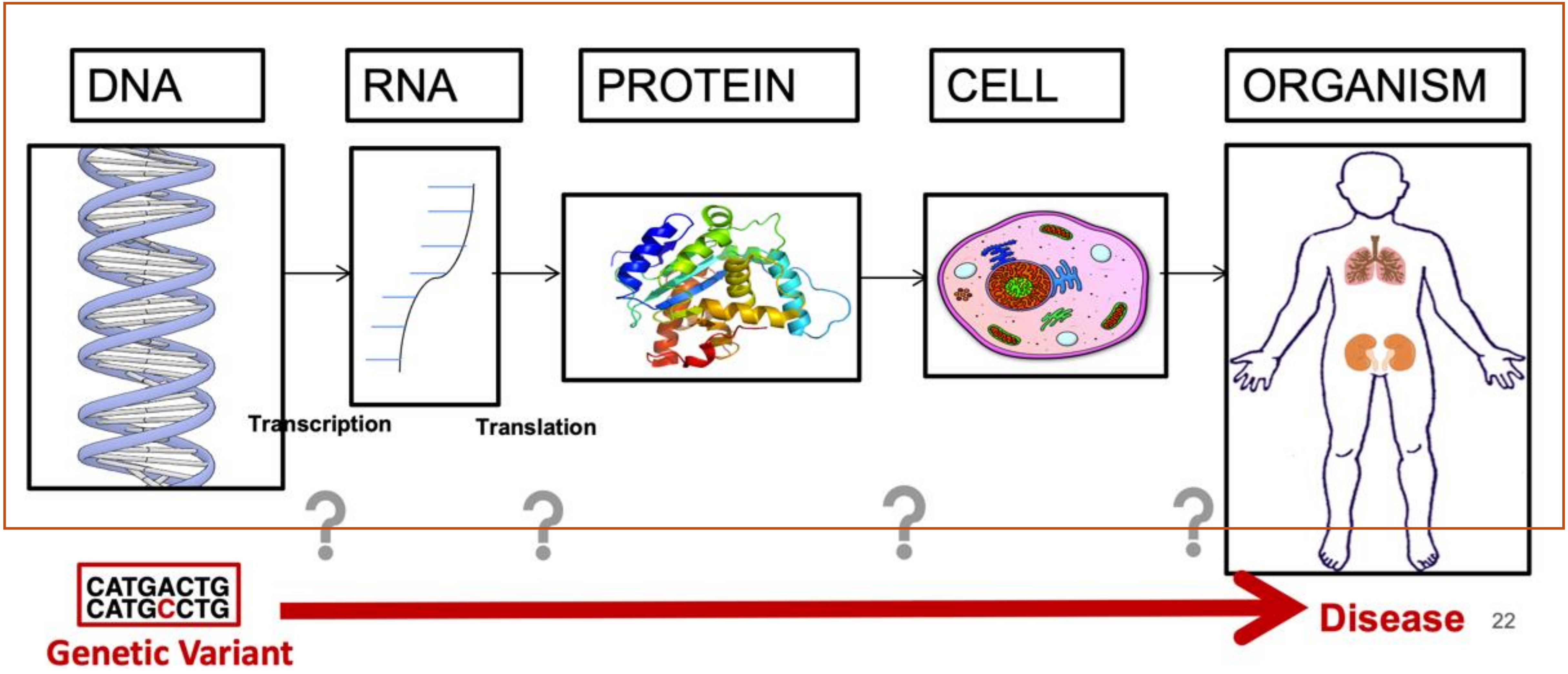
> A model indistinguishable from experiment — a true Turing test for the virtual cell

Summary

Component	Key Detail
Task	Predict gene expression response to genetic perturbations in H1 hESCs (context generalization)
Format	Few-shot: subset of H1 hESC perturbations provided; 100 held-out test perturbations
Dataset	300 CRISPRi perturbations × ~1,000 cells each; 10x Genomics Flex; high depth
Metrics	DE score + perturbation discrimination score + MAE (combined + minimum thresholds)
Training data	Arc Virtual Cell Atlas (>350M cells) + public perturbation datasets
Goal	Reproducible, fair benchmarking to surface generalizable models of cell behavior

Participate: virtualcellchallenge.org | **Data:** arcinstitute.org/tools/virtualcellatlas

Biology in a Slide:



Multimodal AI generates virtual population for tumor microenvironment modeling

Valanarasu et al., *Cell* 189, 386–400, January 22, 2026

Code & model: **https://aka.ms/gigatime_code**

The Tumor Immune Microenvironment (TIME)

- TIME shapes cancer progression, immune evasion, and response to immunotherapy (**Critical — But Hard to Study at Scale**)
- Now: Multiplex immunofluorescence (**mIF**) simultaneously profiles 21 proteins on a single tissue section — the gold standard for TIME analysis
- **Problem:** mIF is expensive, slow, and scarce; most hospitals only generate **routine H&E slides**

💡 **Key gap:** We need mIF-level insight at population scale — but mIF data barely exists outside specialized centers.

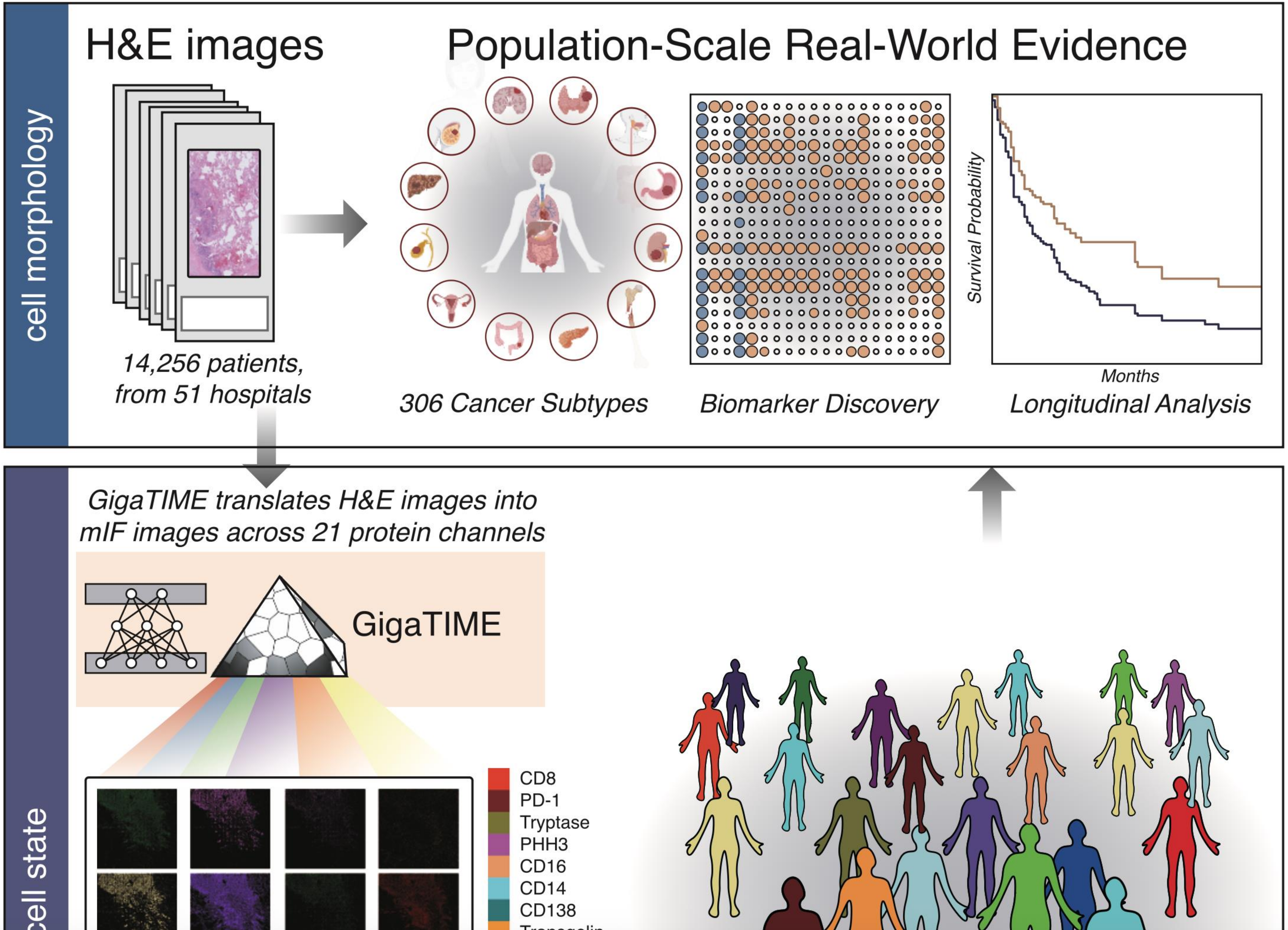
H&E Slides Are Everywhere
— Can AI Read the Hidden Proteomics?

- H&E slides are generated **routinely** in clinical workflows at low cost
- They capture cell morphology and spatial architecture — implicit signals of cell states
- Recent AI foundation models show H&E contains **more information than previously thought**

🎯**Central hypothesis:** Spatial patterns in H&E images are sufficient to predict multiplex protein activation — if we can learn the right cross-modal translation.

Summary of Highlights:

- GigaTIME uses multimodal AI to translate H&E pathology slides to spatial proteomics
- GigaTIME generates a virtual population with cell states from routine H&E slides
- Virtual population enables large-scale clinical discovery and patient stratification
- Virtual population reveals new spatial and combinatorial protein activation patterns


cell morphology
H&E images
Population-Scale Real-World Evidence
14,256 patients,
from 51 hospitals
306 Cancer Subtypes
Biomarker Discovery
Survival Probability
Months
Longitudinal Analysis
cell state
GigaTIME translates H&E images into
mIF images across 21 protein channels
GigaTIME
CD8
PD-1
Tryptase
PHH3
CD16
CD14
CD138

Method: STAR

- We first experimentally acquired 441 mIF images from 21 H&E-stained slides across 21 protein channels (Figure S1; Table 1).
- These paired H&E and mIF slides were processed through a computational pipeline that includes image registration and cell segmentation, resulting in a dataset comprising 40 million cells with paired H&E and mIF slides (Figures 1A and S2).
- We divided the paired data into training, development, and held-out test sets (STAR Methods).

- To translate H&E images into mIF ones, *GigaTIME* was trained on the training paired data using a **patch-based encoder-decoder architecture built on NestedUNet.**

- The model inputs an H&E image patch and outputs corresponding mIF patches, one for each protein channel. These channel-specific patches are subsequently stitched together to reconstruct the whole-slide mIF images, enabling spatially resolved, slide-level protein activation profiling

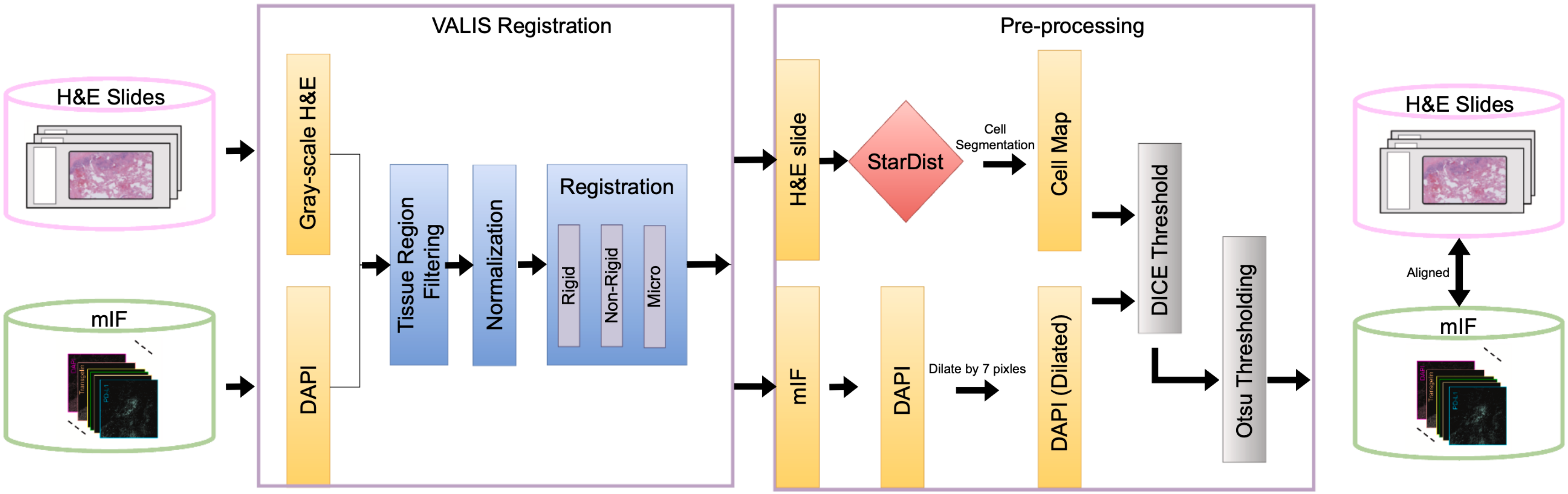


Figure S2. Registration and pre-processing workflow for *GigaTIME* training data, related to Figure 1

Flowchart illustrating the alignment process of H&E and mIF data from the *GigaTIME* training dataset. First, the slides are registered using VALIS. Following registration, pre-processing is performed by computing the Dice threshold similarity between the DAPI channel of the mIF image and the cell map extracted from the H&E slide using StarDist to get the pixel aligned pairs of H&E slides and mIF images.

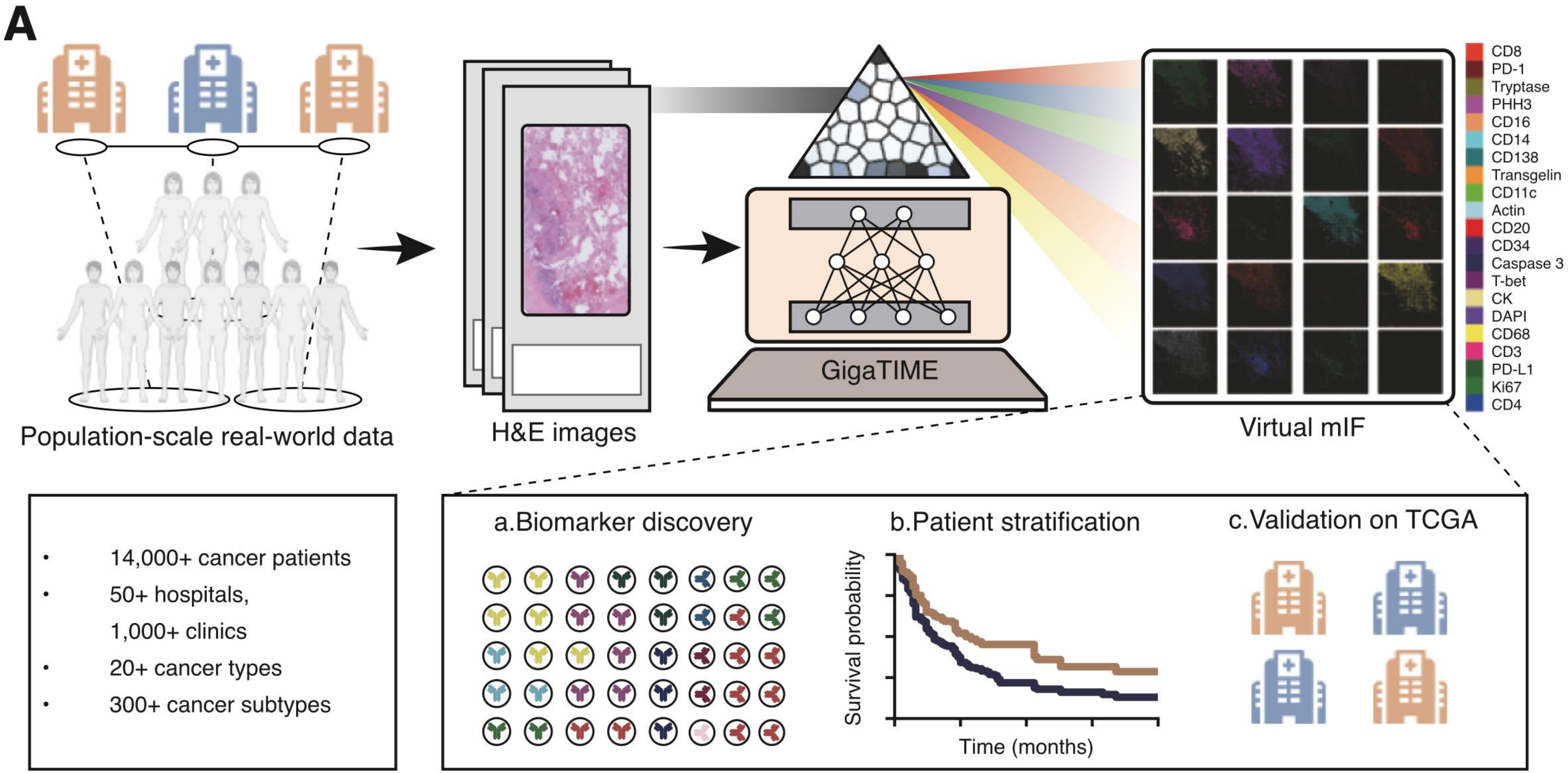
STAR Method Details: GigaTIME Cross-Modal Translator

- **Architecture:** NestedUNet (9.16M parameters) — encoder-decoder with dense nested skip connections
- **Input:** H&E image patch (256×256 pixels)
- **Output:** 21 virtual mIF binary activation maps (one per protein channel)
- **Patch stitching:** Sliding-window inference reconstructs whole-slide virtual mIF
- **Loss:** Combined Dice + Binary Cross-Entropy loss across all 21 channels simultaneously

Basic Motivation: a Scalable Bridge from Morphology to Cell States

- Existing mIF datasets cover **hundreds** of patients at most
- Clinical biomarker discovery requires **thousands** of patients across diverse cancer types
- **A computational approach to generate "virtual mIF" from H&E would unlock population-scale TIME analysis**

Goal: Train a model on paired H&E + mIF data, then apply it to thousands of H&E-only patients to build a *virtual population*


A
Population-scale real-world data
H&E images
GigaTIME
Virtual mIF
CD8
PD-1
Tryptase
PHH3
CD16
CD14
CD138
Transgelin
CD11c
Actin
CD20
CD34
Caspase 3
T-bet
CK
DAPI
CD68
CD3
PD-L1
Ki67
CD4
14,000+ cancer patients
50+ hospitals,
1,000+ clinics
20+ cancer types
300+ cancer subtypes
a.Biomarker discovery
b.Patient stratification
Survival probability
Time (months)
c.Validation on TCGA

Method Details: Training Data — 40 Million Paired Cells

- **21 paired H&E + mIF slides** from lung adenocarcinoma patients (diverse biomarker profiles)
- mIF acquired via **COMET multiplex immunofluorescence** platform (Lunaphore)
- Registration via **VALIS** (multi-step: rigid → non-rigid → micro-registration)
- Quality filtering: Dice threshold on DAPI vs. H&E cell maps → **49,401 high-quality patch pairs**, ~10M cells used for training

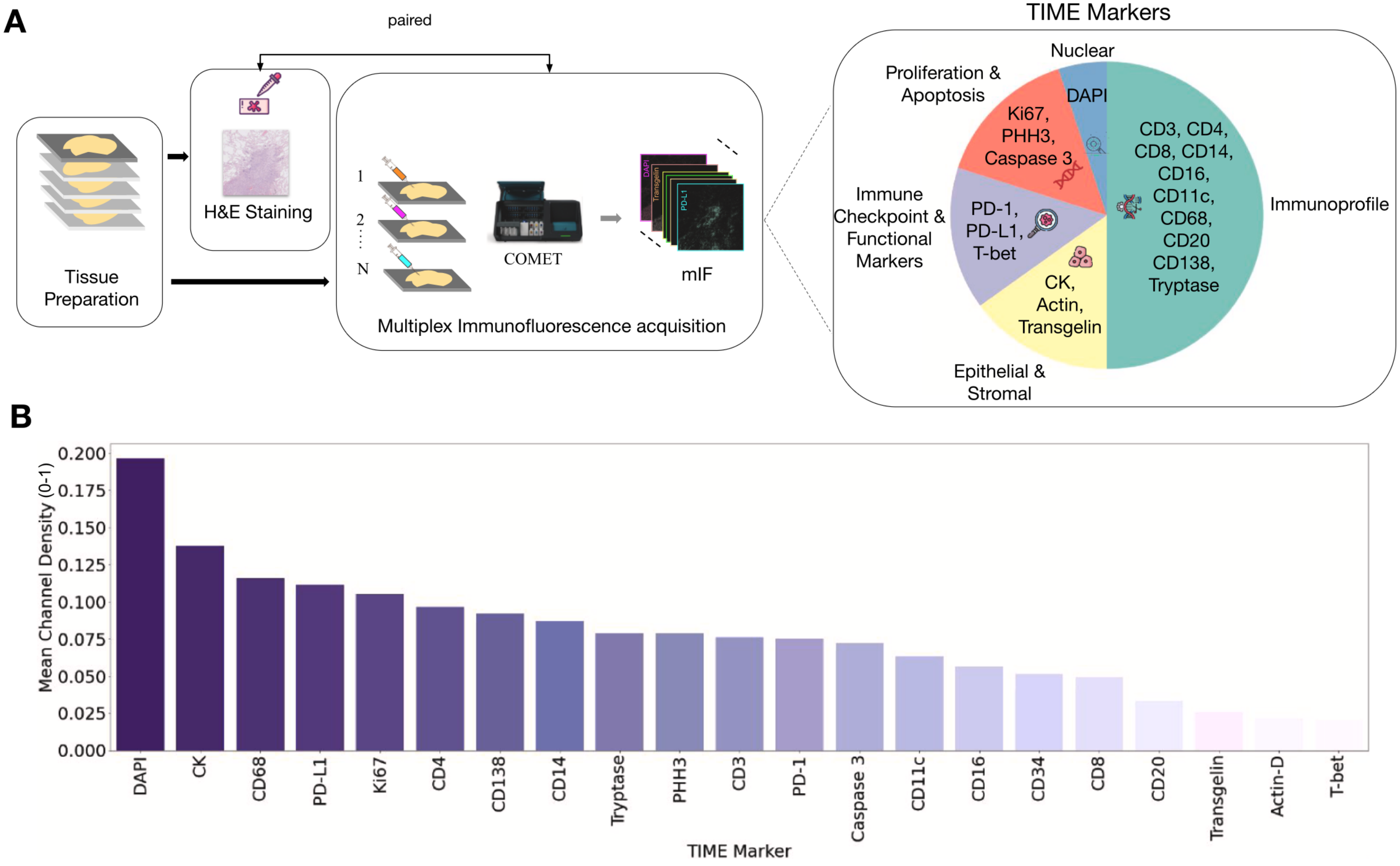


Figure S1. Data acquisition and channel distribution in *GigaTIME* training data, related to Figure 1A

(A) Flowchart illustrating the pipeline for obtaining paired H&E slides and mIF data in the *GigaTIME* training data, including the acquisition process using Lu-

Method Details: 21 TIME Protein Markers Covered

Category	Markers
Proliferation & Apoptosis	Ki67, PHH3, Caspase 3
Immune Checkpoint	PD-1, PD-L1, T-bet
Immunoprofile (T/B/Myeloid)	CD3, CD4, CD8, CD14, CD16, CD11c, CD68, CD20, CD138
Epithelial & Stromal	CK, Actin, Transgelin, Tryptase, CD34, DAPI

Each marker reveals a distinct cell population or functional state in the tumor microenvironment

Baselines: What GigaTIME Is Compared Against

- **CycleGAN** — standard unpaired image-to-image translation model (1 model per channel = 21 models total); widely used for virtual staining
- **Average Activation Baseline** — randomly assigns pixel activations using channel-wise average positivity rate from real mIF data
- **Evaluation metrics** operate at three granularities:
  - **Pixel-level:** Dice score (spatial overlap)
  - **Cell-level:** Pearson correlation of activation counts in 8×8 pixel windows
  - **Slide-level:** Spearman correlation of patch-wise activation density (immunoscore-inspired)

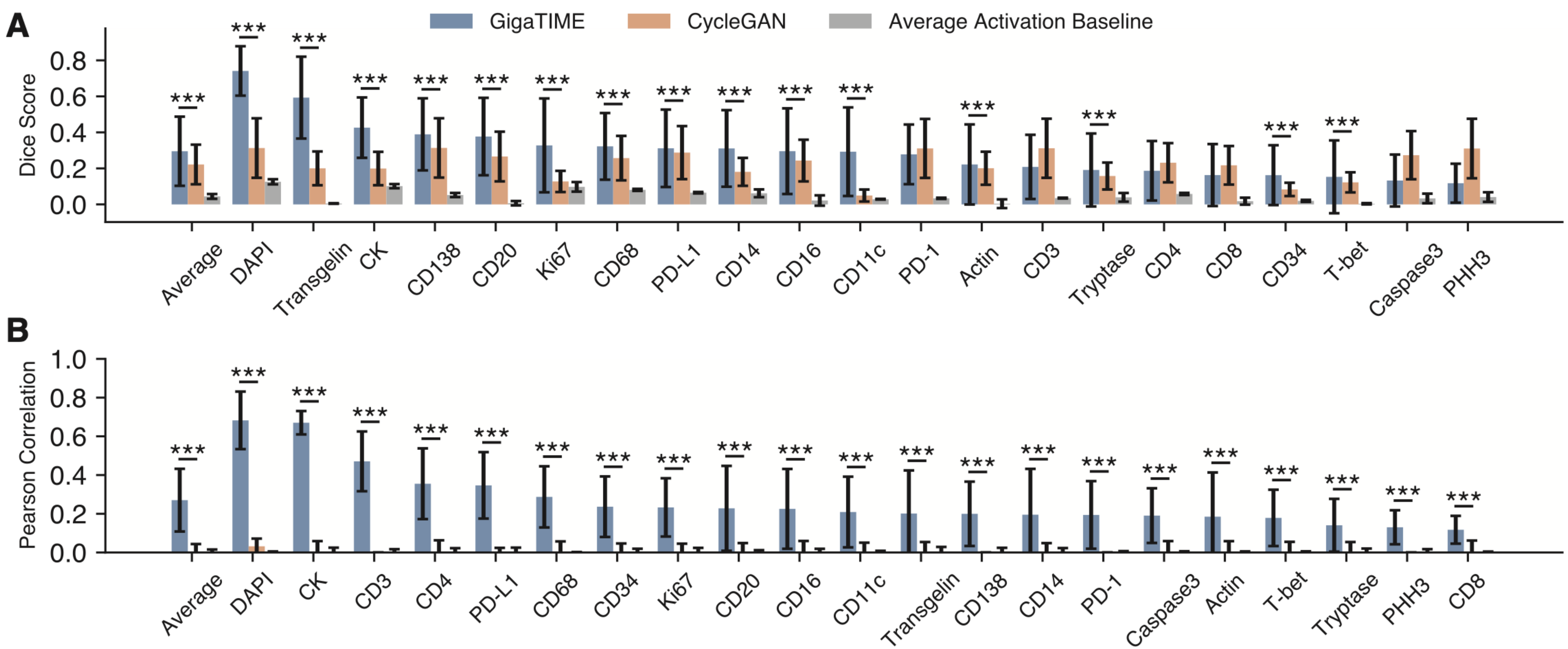
applied it to generate an independent virtual population of 10,200 tumors from the Cancer Genome Atlas (TCGA)

Setup: Training and Inference at Scale

- **Training:** 12 slides (train), 4 (dev), 5 (held-out test); 250 epochs, Adam optimizer (lr=0.0001), 8× A100 GPUs
- **Providence inference:** 14,256 H&E whole-slide images → 299,376 virtual mIF slides across 21 channels; ~5,000 GPU-hours total on 325× V100 GPUs
- **TCGA validation:** 10,200 patients → 214,200 virtual mIF slides
- **Generalization testing:** Additional held-out cohorts — breast TMA and brain TMA (not seen during training)

Insights: GigaTIME Significantly Outperforms CycleGAN

- GigaTIME outperforms CycleGAN on **15/21 channels** (Dice score, pixel level)
- At cell level, GigaTIME Pearson r = 0.59 for DAPI vs. CycleGAN r = 0.03
- At slide level, GigaTIME Spearman r = 0.98 for DAPI, r = 0.56 mean vs. CycleGAN ≈ 0
- Average activation baseline fails entirely (near-zero or negative scores)
- **Nuclear proteins** (DAPI, Ki-67) translate best; cytoplasmic/surface proteins are harder due to diffuse patterns



Takeaways from Methods

- Paired H&E–mIF training data is **essential** — CycleGAN (unpaired) fails to recover cell-level patterns
- A compact 9.16M parameter NestedUNet generalizes well across cancer types and institutions
- The approach scales to population-level: **299,376 virtual slides** generated from routine H&E
- Translation quality varies by protein sub-cellular localization — nuclear > cytoplasmic/surface

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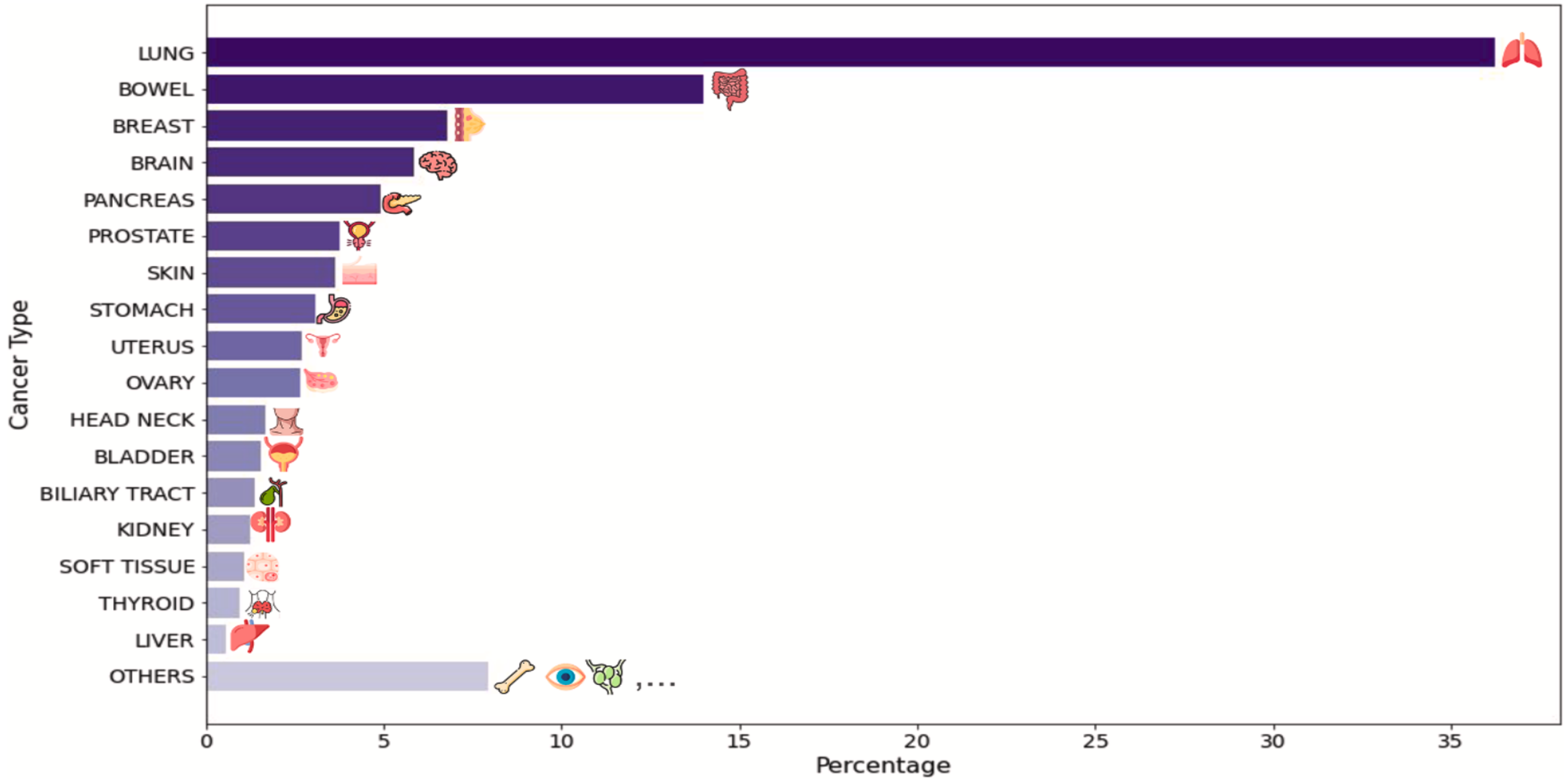
Insights: Generalization to Unseen Cancer Types

- Applied to **breast and brain TMA cohorts** (not in training data)
- Performance ranking across protein channels is broadly conserved
- GigaTIME consistently outperforms CycleGAN and random imputation on unseen cancer types
- TCGA virtual population strongly concordant with Providence (Spearman r = **0.88** at subtype level)

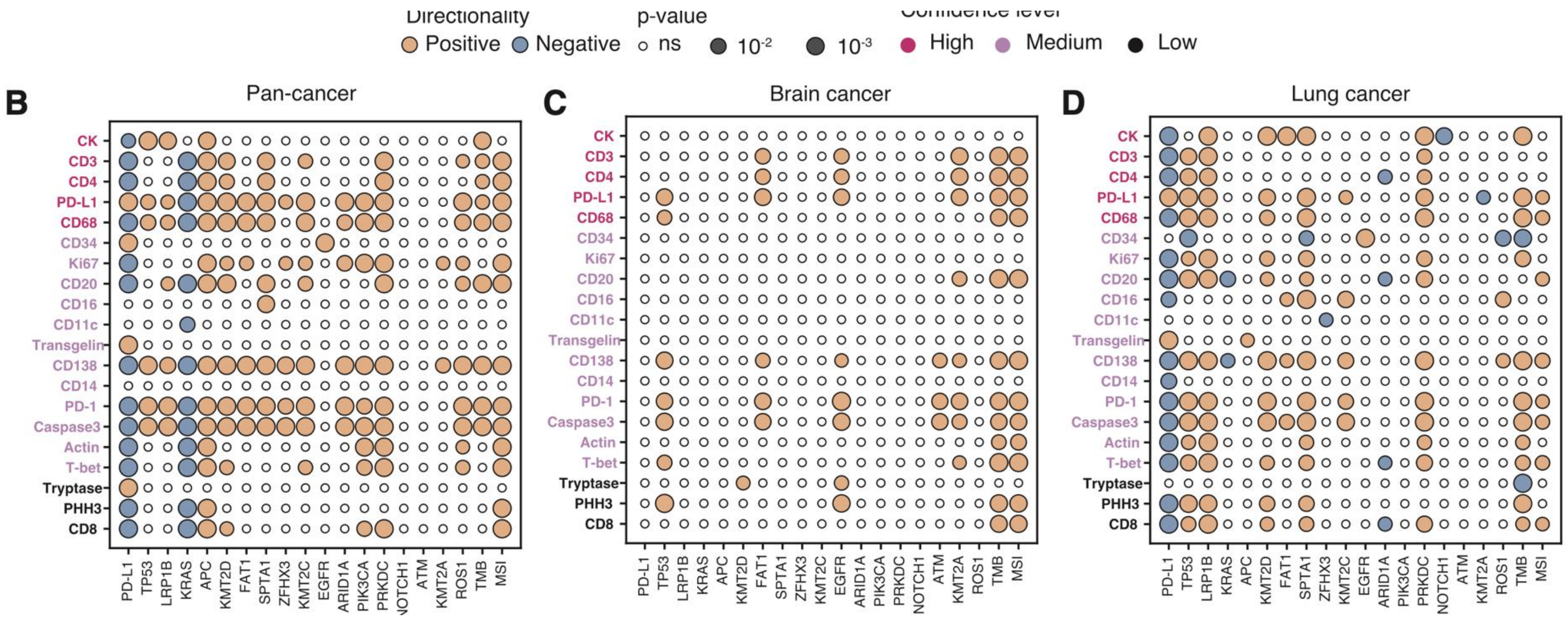
Result 1: A Virtual Population of 14,256 Patients Spans 24 Cancer Types

- Applied to **Providence Health** data: 51 hospitals, 1,000+ clinics, 7 US states
- 14,256 patients → 24 cancer types, **306 cancer subtypes** (OncoTree classification)
- Each patient: virtual mIF across 21 protein channels + clinical attributes (biomarkers, staging, survival)
- Majority of patients: lung, bowel, breast, brain cancers



Result 2: Virtual Population Uncovers 1,234 Protein–Biomarker Associations

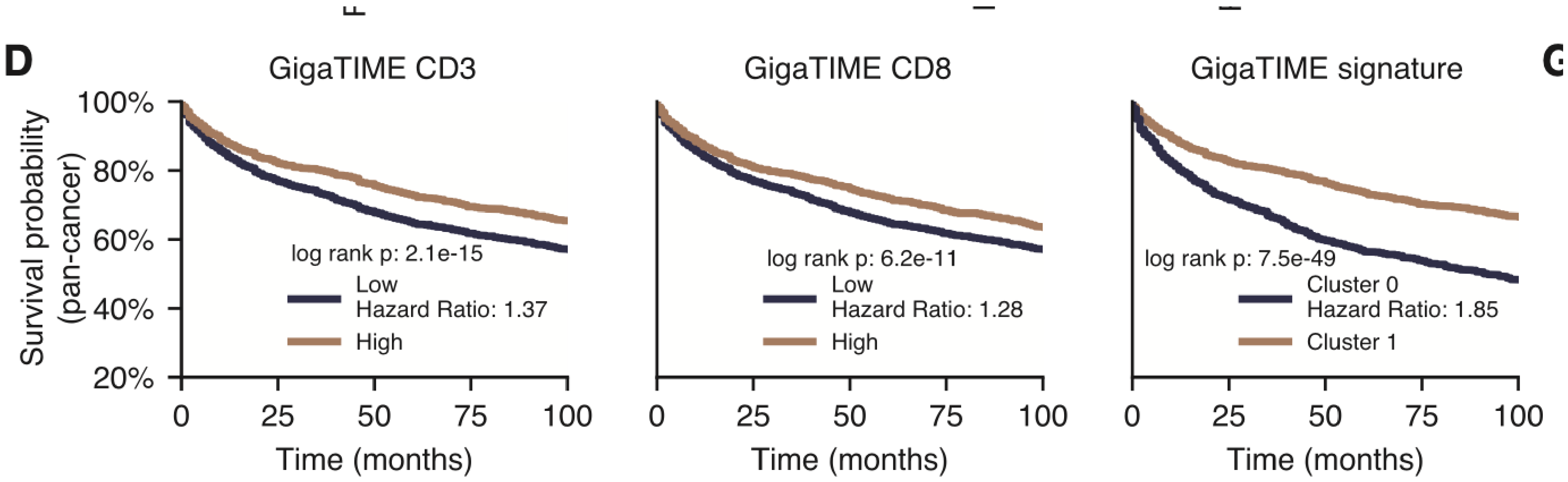
- Tested 21 virtual protein channels against 20 clinical genomic biomarkers (TP53, KRAS, TMB, MSI, PD-L1, etc.)
- **1,234 statistically significant associations** (BH-corrected) across pan-cancer, cancer-type, and subtype levels:
  - Pan-cancer: **175** associations
  - Cancer-type: **453** (brain: 64, lung: 137, bowel: 175)
  - Cancer-subtype: **600**



Known Biology Is Recovered — and New Associations Found

Result 3: Virtual mIF Enables Patient Stratification by Stage and Survival

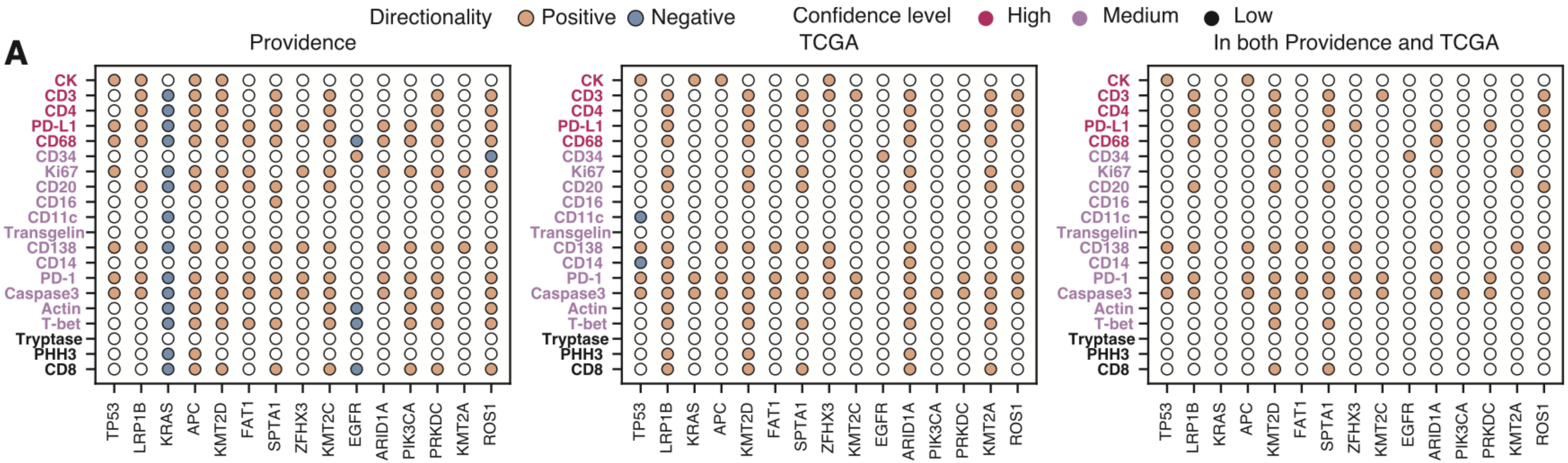
- At pan-cancer level: T stage positively correlates with PD-L1, PD-1, CD68, CD138
- **GigaTIME signature** (all 21 channels combined) outperforms any single protein in survival stratification
- Composite signature log-rank p = **7.5e-49** (pan-cancer); individual channels much weaker
- Integrating all protein channels reveals complementary immune signals missed by single-marker analysis



Survival analysis on lung cancer by using virtual CD3, virtual CD8, and virtual *GigaTIME* signature (all 21 *GigaTIME* protein channels) to stratify patients at a pan-cancer level

Result 4: TCGA Validation Confirms Generalizability

- Independent virtual population generated from **10,200 TCGA patients**
- **80 protein-biomarker associations** significant in both Providence AND TCGA (Fisher's $p < 2\times10^{-9}$)
- Providence yields **33% more** significant pan-cancer associations than TCGA — highlighting value of large, diverse real-world data
- Case studies: patients with SPTA1, KMT2D, TP53 mutations show strikingly different virtual mIF patterns vs. non-mutated patients



GigaTIME Opens Population-Scale TIME Analysis from Routine Pathology

- **What it does:** Translates routine H&E whole-slide images into 21-channel virtual mIF using a multimodal AI trained on 40M paired cells
- **Scale:** 14,256-patient virtual population spanning 306 cancer subtypes — the largest virtual mIF study to date
- **Discoveries:** 1,234 significant protein-biomarker associations, effective patient stratification, novel spatial and combinatorial patterns
- **Validation:** Strong concordance with independent TCGA cohort (r = 0.88)

GigaTIME leverages multimodal AI to generate virtual multiplex immunofluorescence (mIF) profiles from standard H&E slides, enabling comprehensive tumor immune microenvironment modeling across a large (>14,000) and diverse patient population.

This virtual approach unlocks new opportunities for large-scale clinical discoveries that were previously hindered by the scarcity of mIF data.

Limitations and Future Directions

- Most patients from **western US** — geographic and ethnic diversity should be expanded
- **Not all proteins translate equally** — surface/cytoplasmic markers harder than nuclear markers; some proteins may not manifest in H&E morphology
- Training data limited to **lung adenocarcinoma** cases (21 patients)
- Future work: integrate cell segmentation models to study **cell-to-cell interactions** and expand to more protein channels

> 🔗 Code & model weights: **https://aka.ms/gigatime_code**

Recap

- **How to Build the AI Virtual Cell with Artificial Intelligence**
- **Virtual Cell Challenge**
- Multimodal AI generates virtual population for tumor microenvironment modeling